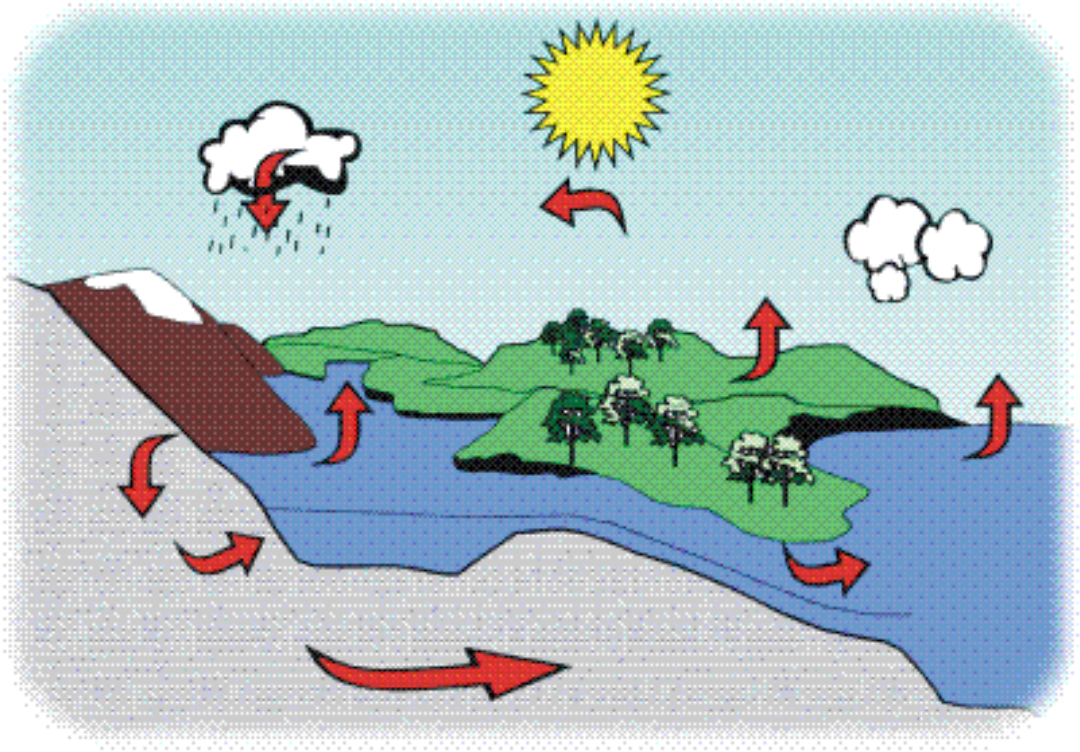


7,8 Losses in Hydrology

Evaporation

Evapotranspiration

Infiltration

Interception

- **The Hydrological Cycle:**

It consists of evaporation, evapo-transpiration, infiltration, interception and depression storage, etc.

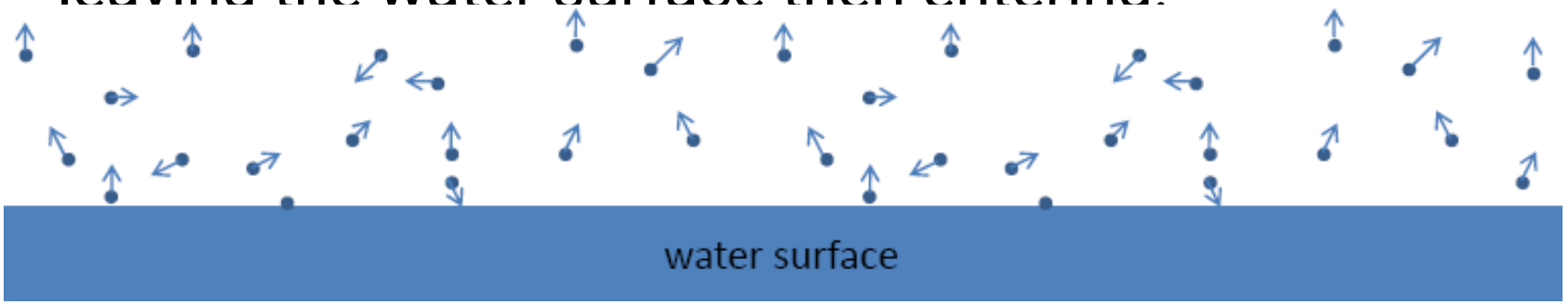
Storage = P-E-ET-IN-INT-DS

- Evaporation and Infiltration are considered as major losses.

Evaporation

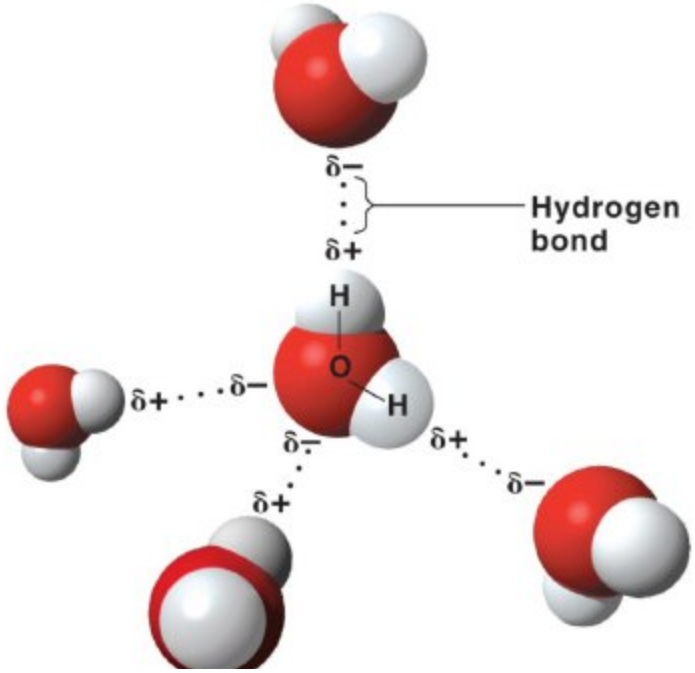
- **Evaporation is the net loss of water from a liquid surface that results from a phase change from a liquid to a vapour.**

- It is a net process because water vapours are constantly moving back and forth from the water surface.
- Evaporation occurs when there are more molecules leaving the water surface then entering.

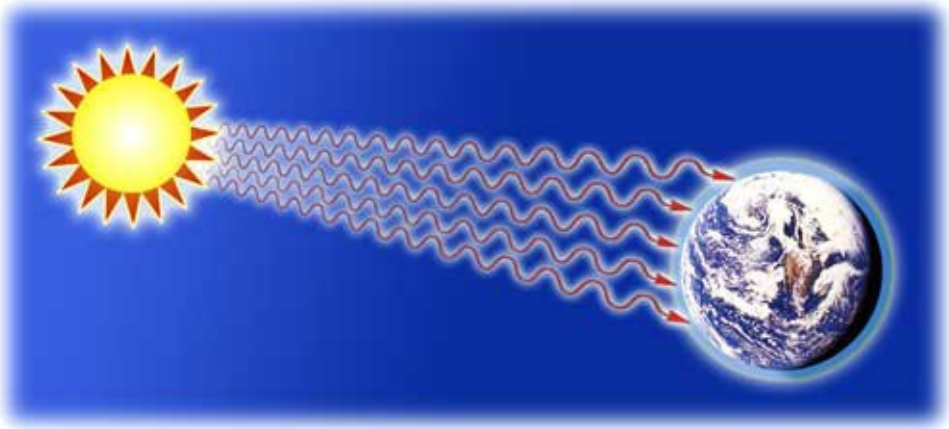


Evaporation process

- Liquid water is a collection of molecules connected by hydrogen bonds.

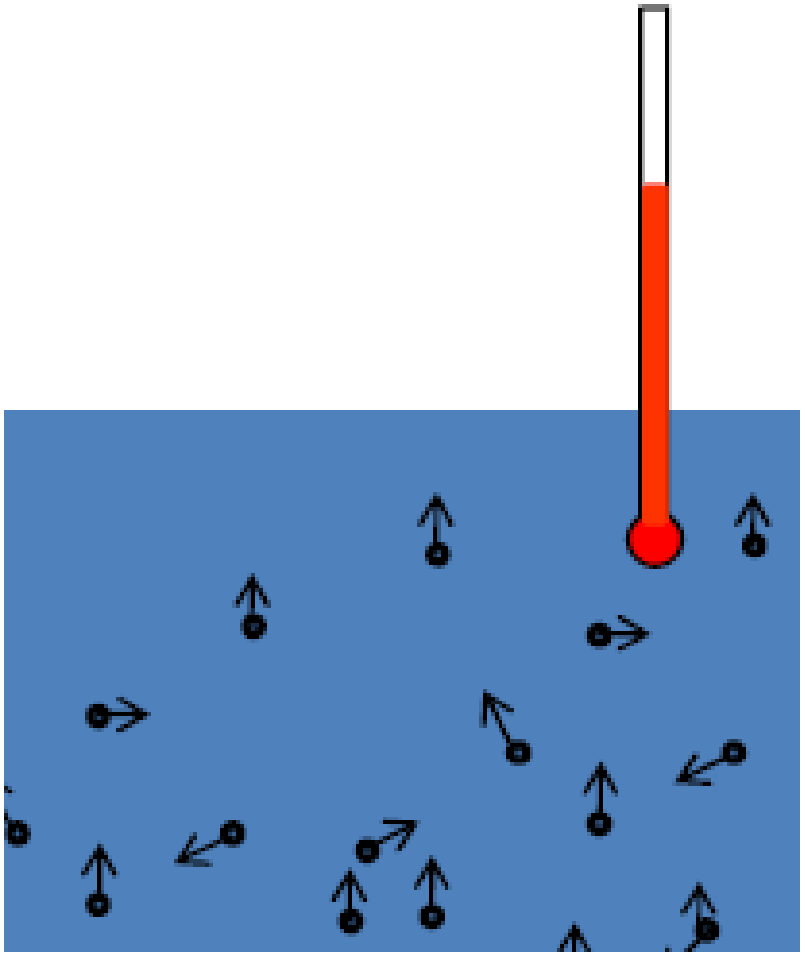


- It takes energy for a molecule to break a hydrogen bond and escape the water surface.

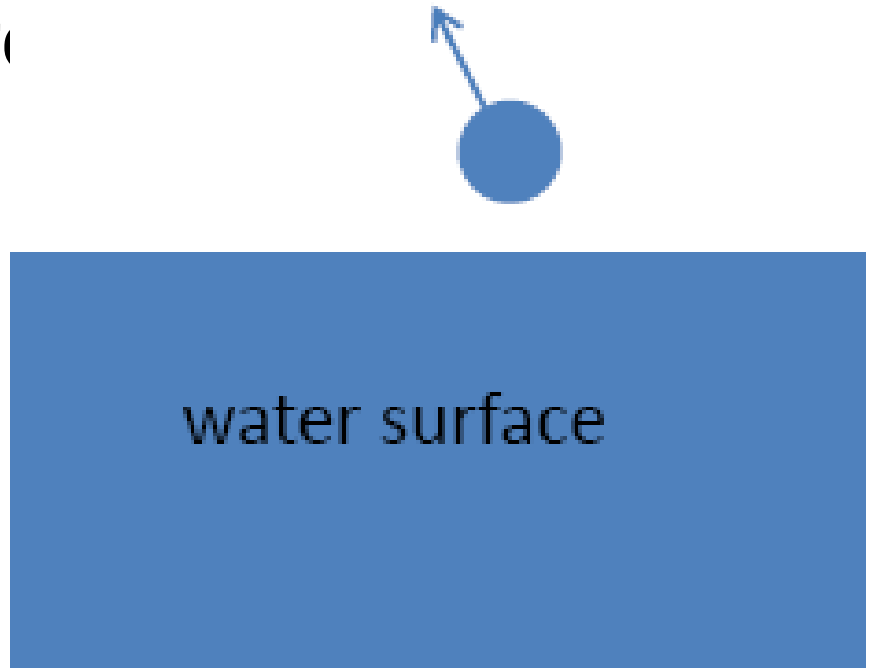


- The collective motion of water molecules in the water is called kinetic energy.

- The average kinetic energy of the molecules quantifies the water temperature.



- Molecules with the highest kinetic energy can break the hydrogen bonds and escape the water surface.
- Thus, reducing the water temperature because the average kinetic energy is reduced (heat is r



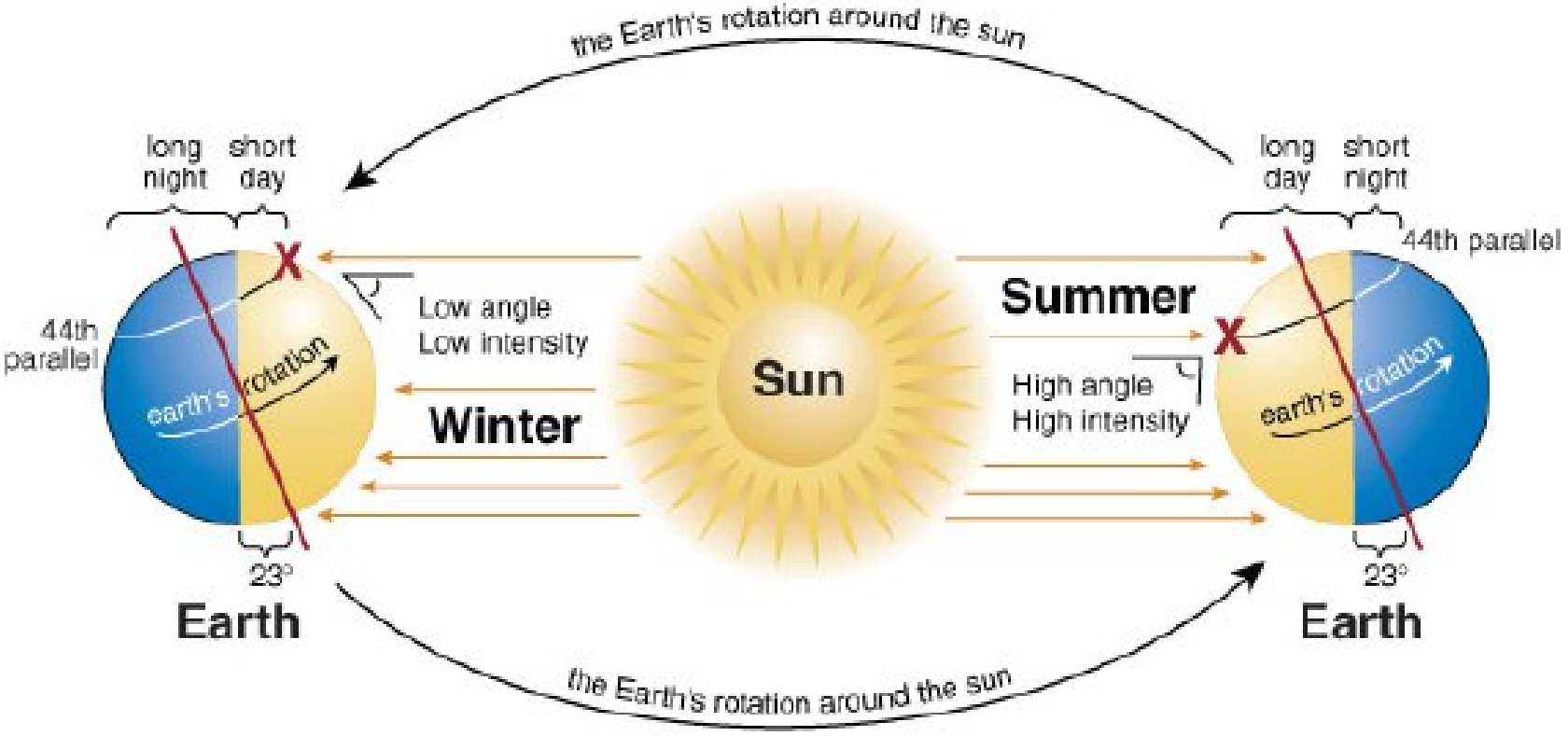
- What drives evaporation?

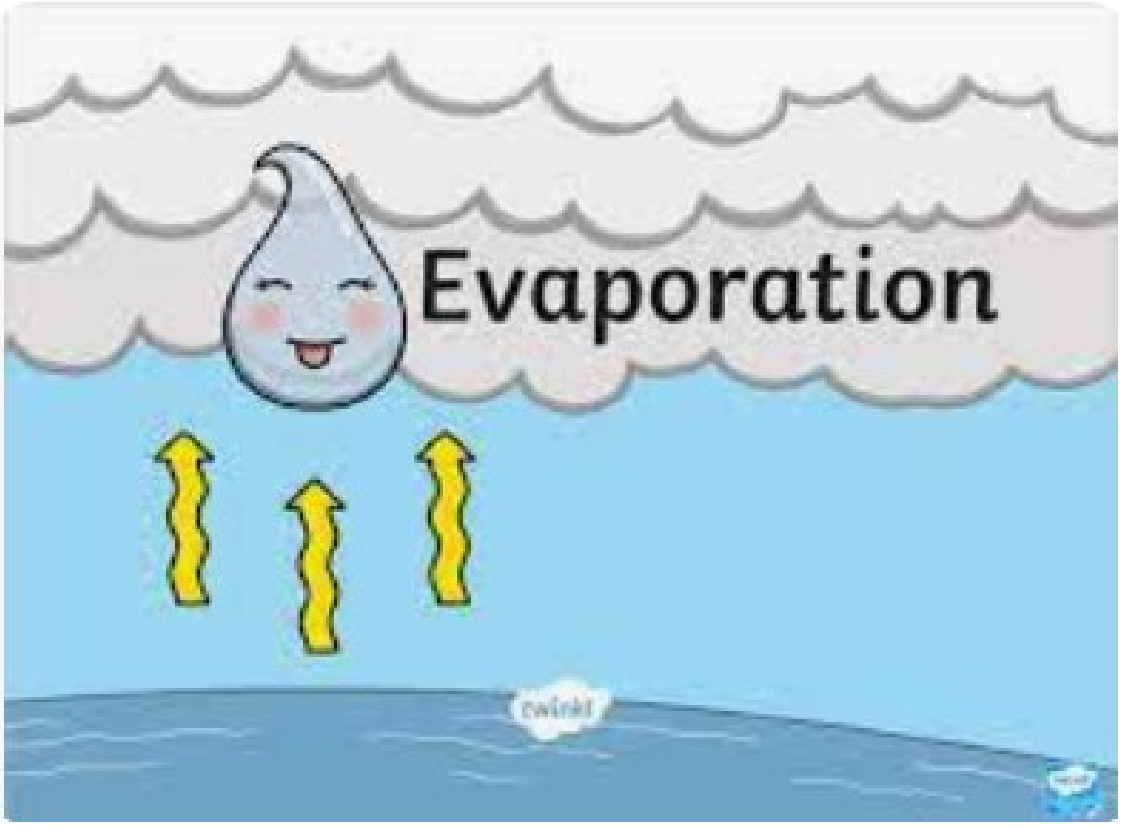
  Energy input to the water to supply heat (i.e., increase the kinetic energy so molecules can escape).

  Diffusion of water vapour molecules from the water surface to the atmosphere.

  Transport of water vapour molecules away from the water surface.

- **Solar radiation** is the main source of heat to a lake.
- The amount of solar radiation depends on the **time** of year and **latitude.**





Factors affecting evaporation

- **Temperature:**
  - Evaporation is highly dependent upon temperature. Higher the temperature, greater will be the evaporation.
- **Surface area of water body:**
  - Evaporation takes place from free surface of water body. Larger the area exposed to atmosphere, greater will be evaporation.
- **Humidity:**
  - Greater the humidity, lesser will be the evaporation

- **Wind speed:** Higher the wind speed more will be evaporation.
- **Solar radiation:** Evaporation increases with increasing solar radiation.
- **Atmospheric Pressure:** Evaporation decreases with increasing atmospheric pressure.
- **Depth of Water:** Greater the depth of water, lesser will be the evaporation.



Measurement/Estimation

- Evaporation can be estimated by:
  - Indirect or theoretical method
    - Water Budget Approach,
    - Energy Budget Approach, and
    - Mass Transfer Approach
  - Direct measurement using pans etc

Estimation of Evaporation

1. Water Budget Approach

- The water budget approach is the least complicated method of estimating evaporation but results obtained from it are not so accurate. It involves writing the hydrological continuity equation for the lake and determining the evaporation from a knowledge or estimation of others variables. Thus considering the daily average values for a lake, the continuity equation is written as

Continued....

Estimation of Evaporation

- $\mathbf{P + V_{is} + V_{ig} = V_{os} + V_{og} + E_L + \Delta S + T_L}$
  - Where
  - **P** = daily precipitation
  - $\mathbf{V_{is}}$ = daily surface inflow into the lake
  - $\mathbf{V_{ig}}$ = daily ground water inflow
  - $\mathbf{V_{os}}$ = daily surface outflow from the lake
  - $\mathbf{V_{og}}$ = daily seepage outflow
  - $\mathbf{E_L}$ = daily lake evaporation
  - $\mathbf{\Delta S}$ = increase in lake storage in a day
  - $\mathbf{T_L}$ = daily transpiration loss

Continued....

Estimation of Evaporation

1. Water Budget Approach

- From the above equation the following equation can be deduced

$$E_L = P + (V_{is} - V_{os}) + (V_{ig} - V_{og}) - T_L - \Delta S$$

Continued....

Estimation of Evaporation

2. Energy Budget Approach

The energy budget approach makes use of the law of conservation of energy which indicates that the total energy reaching a body of water must be equal to the total energy leaving the water plus the increase in the internal energy of the water body over a known time interval.

Continued....

Estimation of Evaporation

Considering the water body shown, the energy balance to the evaporating surface in a period of one day is given by.

$$H_n = H_a + H_e + H_g + H_s + H_i$$

Where,

H_n = net heat energy received by the water surface

$= H_c(1-r) - H_b$

in which $H_c(1-r)$ = incoming solar radiation into a surface of reflection coefficient (albedo), r.

Continued....

Estimation of Evaporation

H_a = Sensible heat transfer from water surface to air

H_b = Back radiation(long wave) from water body

H_e = Heat energy used up in evaporation

= $\rho L E_L$ where

ρ = Density of water,

L = Latent heat of evaporation and

E_L = Evaporation in mm

H_g = Heat flux into the ground

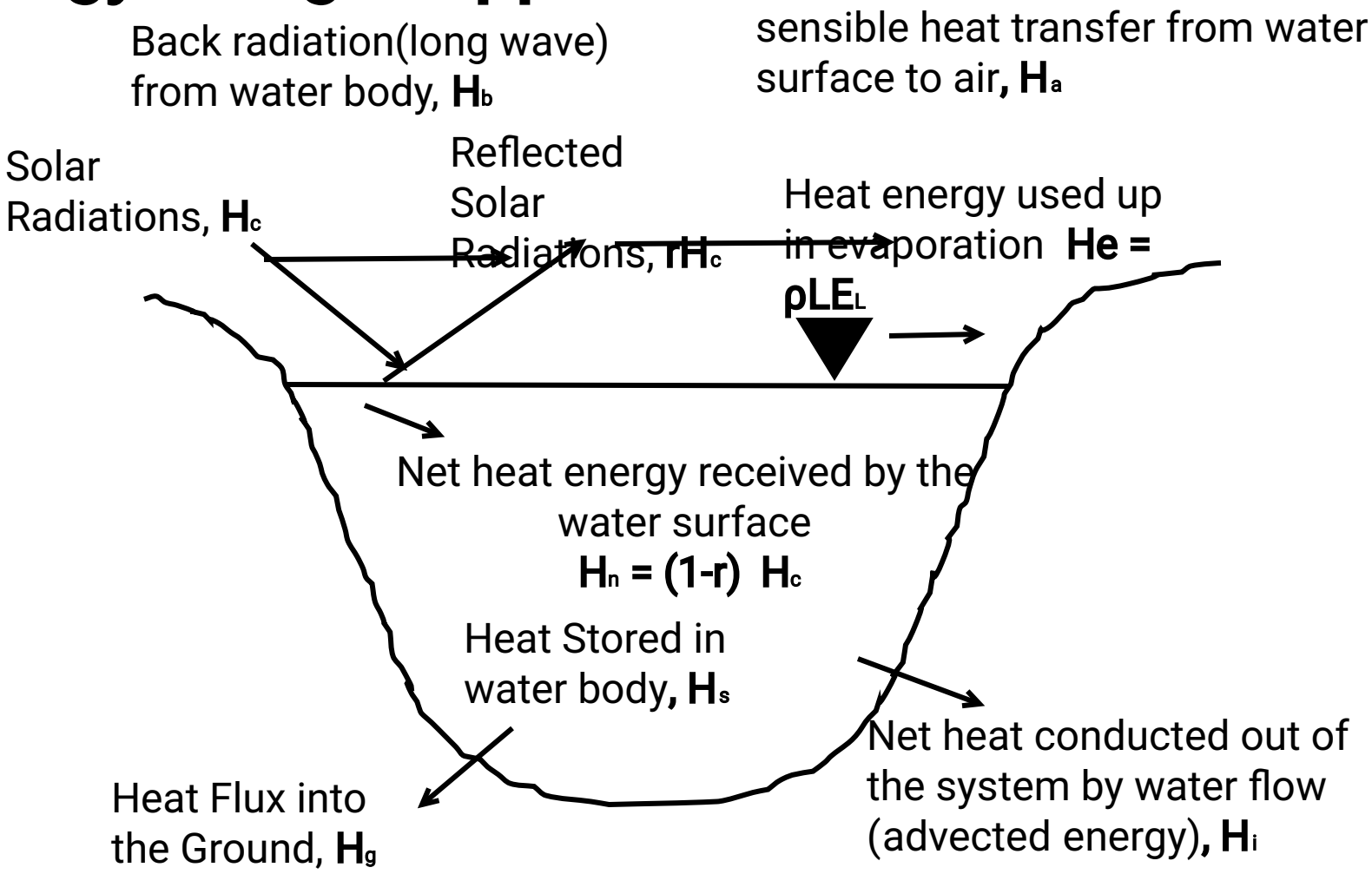
H_s = Heat stored in water body

H_i = Net heat conducted out of the system by water flow (advected energy)

Continued....

Estimation of Evaporation

2. Energy Budget Approach



Continued....

Estimation of Evaporation

- The sensible heat term H_a which cannot be readily measured is estimated using Bowen's Ratio β given by the expression as

$$\beta = H_a / H_e = H_a / (\rho L E_L) = 6.1 \times 10^{-4} \times p_a [(T_w - T_a) / (e_w - e_a)]$$

- Where

p_a =atmospheric pressure in mm of mercury

e_w = saturated vapor pressure in mm of mercury

e_a = actual vapor pressure of air in mm of mercury

T_w = temperature of water surface in ^{0}C and

Continued....

Estimation of Evaporation

From this E_L can be evaluated as

$$E_L = [(H_n - H_g - H_s - H_i) / \rho L(1 + \beta)]$$

This is the expression by the help of which the evaporation can be estimated

Continued....

Estimation of Evaporation

3. Mass Transfer Approach

The mass transfer approach contains Dalton's original idea that evaporation ratio is determined by a combination of influences including wind, humidity, sunshine and surface characteristics.

His equation is given below:

$$E = C (e_o - e_a)$$

Continued....

Estimation of Evaporation

- In this equation the evaporation ‘E’ is a product of the vapor pressure gradient existing between the evaporating surface ‘e_o’ and the air above the surface ‘e_a’ and a constant including wind and empirical functions given by following equation:

$$C = 0.622\, k^2\, \rho_a\, v \,/(\, p\, \rho_w\, [\ln (z/z_o)]^2)$$

Continued….

Estimation of Evaporation

- Where
  - k = Von Karman constant = 0.4
  - ρ_a = Density of air
  - v = Velocity of wind at elevation z above the water surface
  - p = Air pressure
  - ρ_w = Density of water
  - z = Elevation at which ‘v’ is measured
  - z_o = Height of roughness.

Continued….

- The 'roughness length' z_o, depends on the frontal area of the average element (facing the wind) divided by the ground width it occupies

Class No.	Class Name	Roughness length: m	Landscape Features
1	Sea	0.0002	open water, tidal flat, snow with fetch above 3 km
2	Smooth	0.005	featureless land, ice
3	Open	0.03	flat terrain with grass or very low vegetation, airport runway
4	Roughly open	0.10	cultivated area, low crops, obstacles of height H separated by at least 20 H
5	Rough	0.25	open landscape, scattered shelter belts, obstacles separated by 15 H or so
6	Very rough	0.5	landscape with bushes, young dense forest etc separated by 10 H or so
7	Closed	1.0	open spaces comparable with H, eg mature forest, low-rise built-up area
8	Chaotic	over 2.0	irregular distribution of large elements, eg city centre, large forest with clearings

Mass Transfer Approach

- Example 3
  - Calculate evaporation from an open water surface in a region where air pressure is 105 KN/m². The vapor pressure of water is 3.167 KN/m² and that of air is 1.583 KN/m². The wind speed measured at elevation of 1.75 m above water surface is 2.7 m/s. Assume height of surface roughness as 0.3 mm. The density of water and air may be taken as 1,000 kg/m³ and 1.2 kg/m³ and the Von Karman constant is 0.40.

Continued....

Estimation of Evaporation

Solution:

The given data is arranged below with consistent units:

$k = 0.4$

$\rho_a = 1.2\ kg/m^3$

$v = 2.7\ m/s$

$p = 105 \times 10^3\ N/m^2$

$\rho_w = 1000\ kg/m^3$

$z = 1.75\ m$

$z_o = 0.0003\ m$

$e_o = 3167\ N/m^2$

$e_a = 1583\ N/m^2$

Substituting known data in Equation A and B, gives:

$E = 4.16 \times 10\text{-}11\ m/s = 5.7\ mm/day$

Methods for Measuring Evaporation

- Because of many difficulties encountered in the direct measurement of lake-evaporation using water budget or energy budget methods, direct methods of estimation are used. The estimation is done on the basis of evaporation from a **pan**.

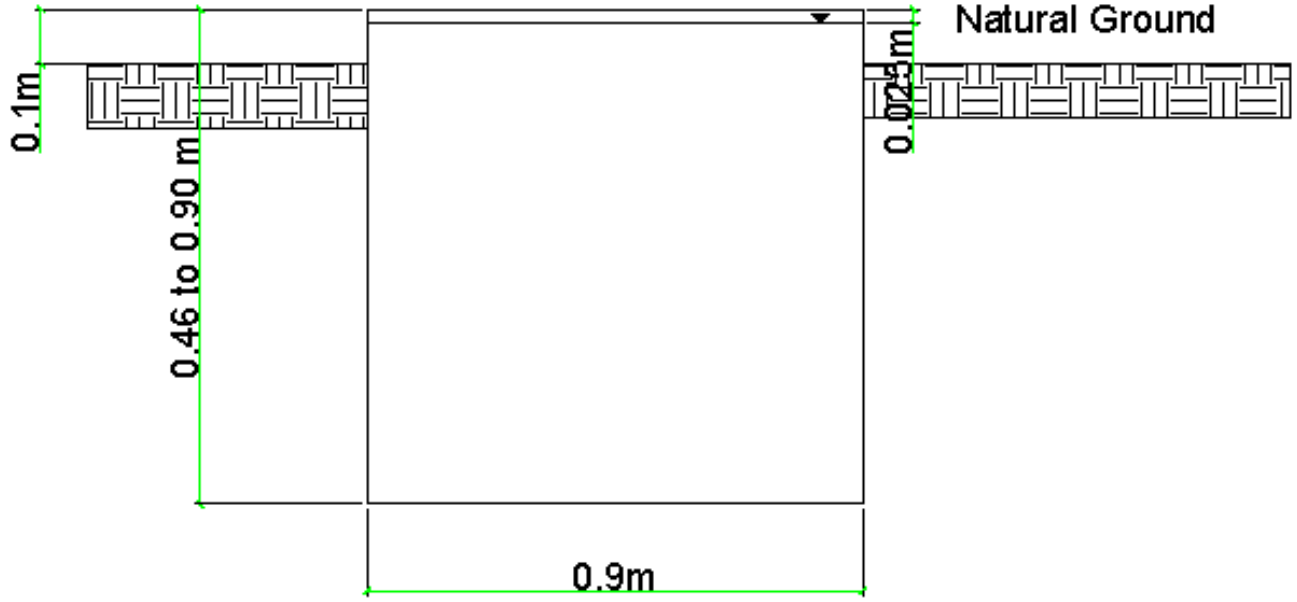
- Evaporation Pans
  - The pan is the most widely used evaporation instrument. There are three types of pans.
    - **Sunken pans**

Continued....

Methods for Measuring Evaporation

Sunken Pan

This is installed inside the earth surface. This pan tends to eliminate objectionable boundary effects such as radiation on the side walls and heat exchange between the atmosphere and the pan. The results obtained from this pan are more accurate than those obtained from other pans.



Continued....

Sunken Pans



Methods for Measuring Evaporation

- Sunken pans are of three types.

1. Bureau of Plant Industry (B. P. I.) Pan

This pan is circular in shape. These are 1.80 m in diameter and 0.60 m deep. These are sunken into the ground to a depth of 0.55 m and 0.05 m above the ground i.e. total depth is 0.60 m.

2. Young's Pan

It is also circular in shape, about 0.60 m in diameter and 0.90 m deep covered with a 6 mm wire mesh screen. Other conditions are same as that of Colorado pan.

3. Colorado Sunken Pan

- This is 92 cm square and 42-92 cm deep and is sunk in the ground such that only 5-15 cm depth projects above the ground surface and thus the water level is maintained almost at the ground level.

- The evaporation is measured by a point gauge.

Methods for measuring Evaporation

- Draw Backs
  - It is difficult to clean the pan
  - If there is any leakage it cannot be easily detected and rectified.
  - The height of the vegetation adjacent to the pan is quite critical.
  - As it is open and just 5-10 cm above the ground surface so seaweeds, leaves, garbage, debris etc. can easily enter into the pan.

Continued....

Above Ground or Surface Pan

- The surface pan which has found international recognition is U.S. Weather Bureau class-A pan.
- This is made of galvanized iron (G.I.) sheet.
- The pan is placed on a wooden frame to allow free access of air at the bottom (see figure).
- The principle advantage of the surface pan is the easy operation and

Continued....

Above Ground or Surface Pan

- However more evaporation takes place as compared to the sunken and floating pan but this type of pans are most widely used all over the world. Certain correction factors are applied to every pan.

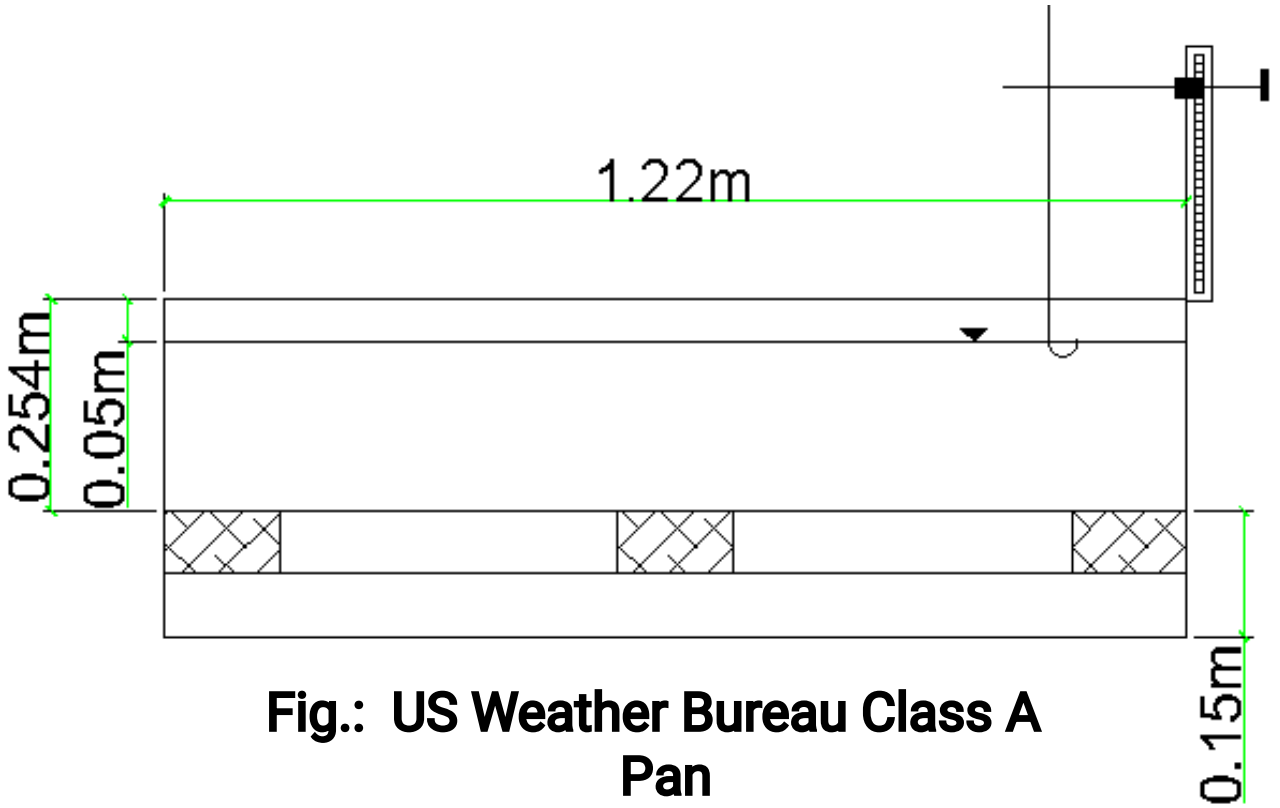
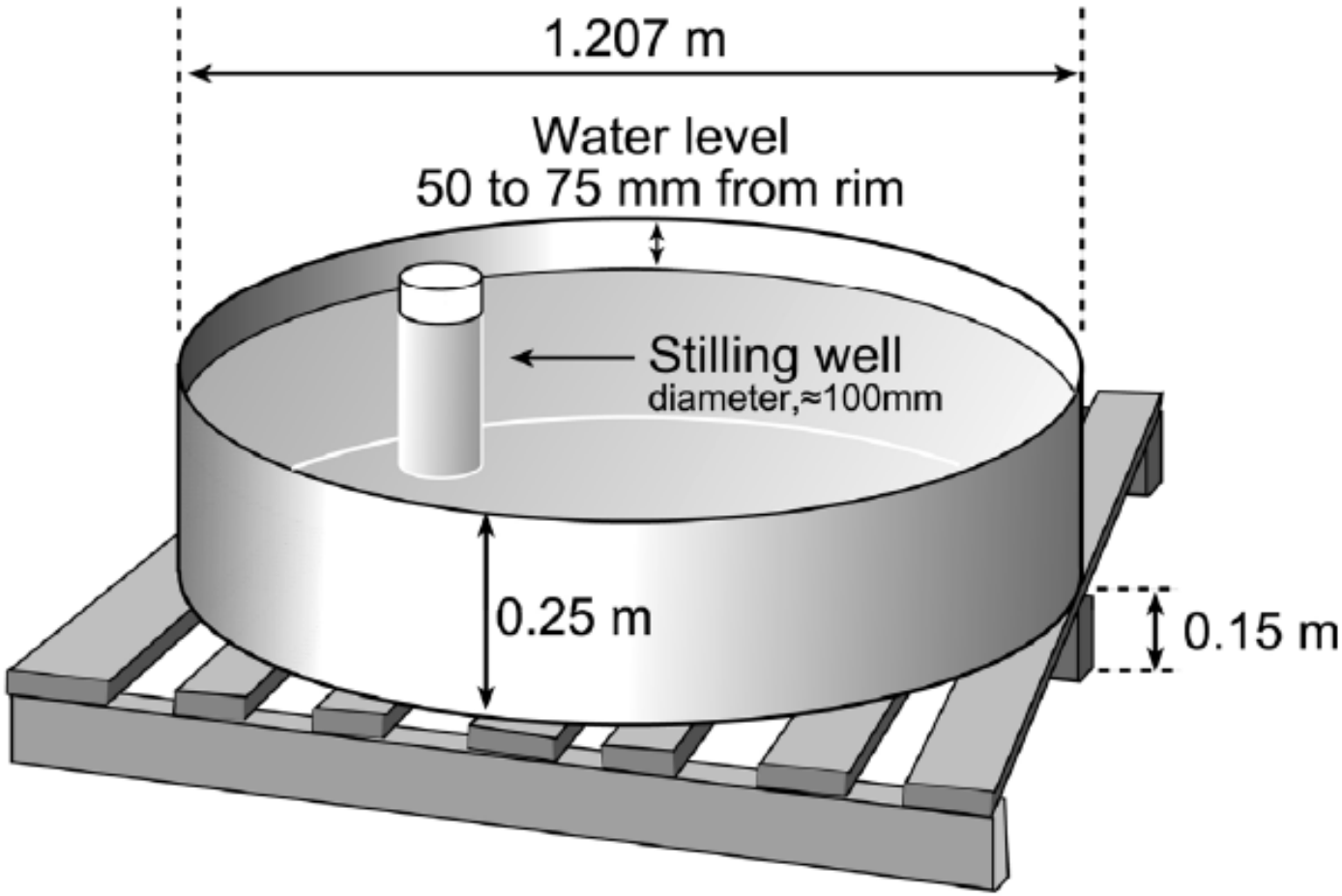


Fig.: US Weather Bureau Class A Pan

- Above Ground Pan

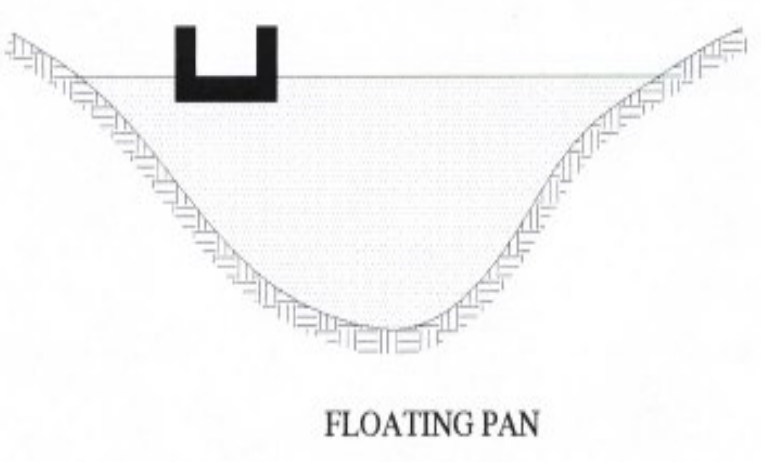




- The pan has a stilling well, vernier point gauge, a thermometer with clip and may be covered with a wire screen.

- The amount of water lost by evaporation from the pan can be directly measured by the point gauge.

Floating Pans





- As the name indicates the floating pans float over the surface of lake. With a certain arrangement the pan is installed in the lake so that it could float over the surface of lake.
- Evaporation from a floating pan is approximately the same as that directly from the lake, because the atmospheric conditions around the pan are the same as those for the lake.

- Allowance has to be made for rainfall, if there has been any. Water is added to the pan from a graduated cylinder to bring the water level to the original mark, i.e., 5 cm below the top of the pan.
- Experiments have shown that the unscreened pan evaporation is 1.144 times that of the screened one.

Draw Backs

- The pans do not give the exact measure of large reservoir with following drawbacks
  - They differ in heat storing capacity.
  - Heat transferred from sides and bottom.
  - Height of rim in evaporation pan affects the wind action over the surface.
  - The rim height also casts a shadow of variable magnitude over surface of water.
  - Heat transfer characteristics of pan material is different from that of the reservoir.
  - Due to these reasons a correction factor is required.

Pan Coefficient

- The correction factor is given as
- Lake Evaporation = C_p x Pan evaporation
- Pan coefficient values are given as

Sr. No	Type of Pan	Average Value	Range
1	Class A Land type	0.70	0.60-0.80
2	ISI Pan (Modified Class A)	0.80	0.65-1.1
3	Coloroda Sunken Pan	0.78	0.75-0.86
4	USGS Floating Pan	0.80	0.70-0.82

- Compute the daily evaporation from a Class A pan if the amounts of water added to bring the level to the fixed point are as follows:

Day:	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Rainfall (mm):	*14*	*6*	*12*	*8*	*0*	*5*	*6*
Water added (mm):	*–5 (removed)*	*3*	*0*	*0*	*7*	*4*	*3*

What is the evaporation loss of water in this week from a lake (surface area = 640 ha) in the vicinity, assuming a pan coefficient of 0.75?

Sol:

Day:	*1*	*2*	*3*	*4*	*5*	*6*	*7*
Rainfall (mm):	*14*	*6*	*12*	*8*	*0*	*5*	*6*
Water added (mm):	*–5 (removed)*	*3*	*0*	*0*	*7*	*4*	*3*

$$\text{Pan evaporation, } E_p, \text{ mm} = \text{Rainfall} \begin{matrix} + \text{ water added} \\ \text{or} \\ - \text{ water removed} \end{matrix}$$

Day:	1	2	3	4	5	6	7
E_p (*mm*):	14 – 5 = 9	6 + 3 = 9	12	8	7	5 + 4 = 9	6 + 3 = 9

Pan evaporation in the week $= \sum_{1}^{7} E_p = 63$ mm

Pan coefficient $0.75 = \dfrac{E_L}{E_p}$

$\therefore$ Lake evaporation during the week E_L

$= 63 \times 0.75 = 47.25$ mm

Water lost from the lake $= A \,.\, E_L = 640 \times \dfrac{47.25}{1000} = 30.24$ ha.m

- **Transpiration** is the process by which water leaves the body of a plant and reaches the atmosphere as water vapor. The water is taken by plant root system and escapes trough the leaves.

- Factors effecting the transpiration are
  - Atmospheric vapor pressure
  - Temperature
  - Wind
  - Light intensity and characteristics of plant

- Transpiration is confined to day light hours and the rate of transpiration depends upon the growth periods of plants
- Evaporation on the other hand continues all through day and night

- The evaporation and transpiration processes can be considered under one head of evapo-transpiration, ET.

- Consumptive use is also used to denote this loss by evapo-transpiration. Evapo-transpiration depends on the available water

- For a given vegetation type measurement of evapotranspiration can be done by Lysimeters.

- **Lysimeters** are special water tight tanks containing a block of soil planted with same vegetation as nearby and set in a field of growing plants.

- Evaporation is estimated in terms of amount of water required to maintain

Lysimeter





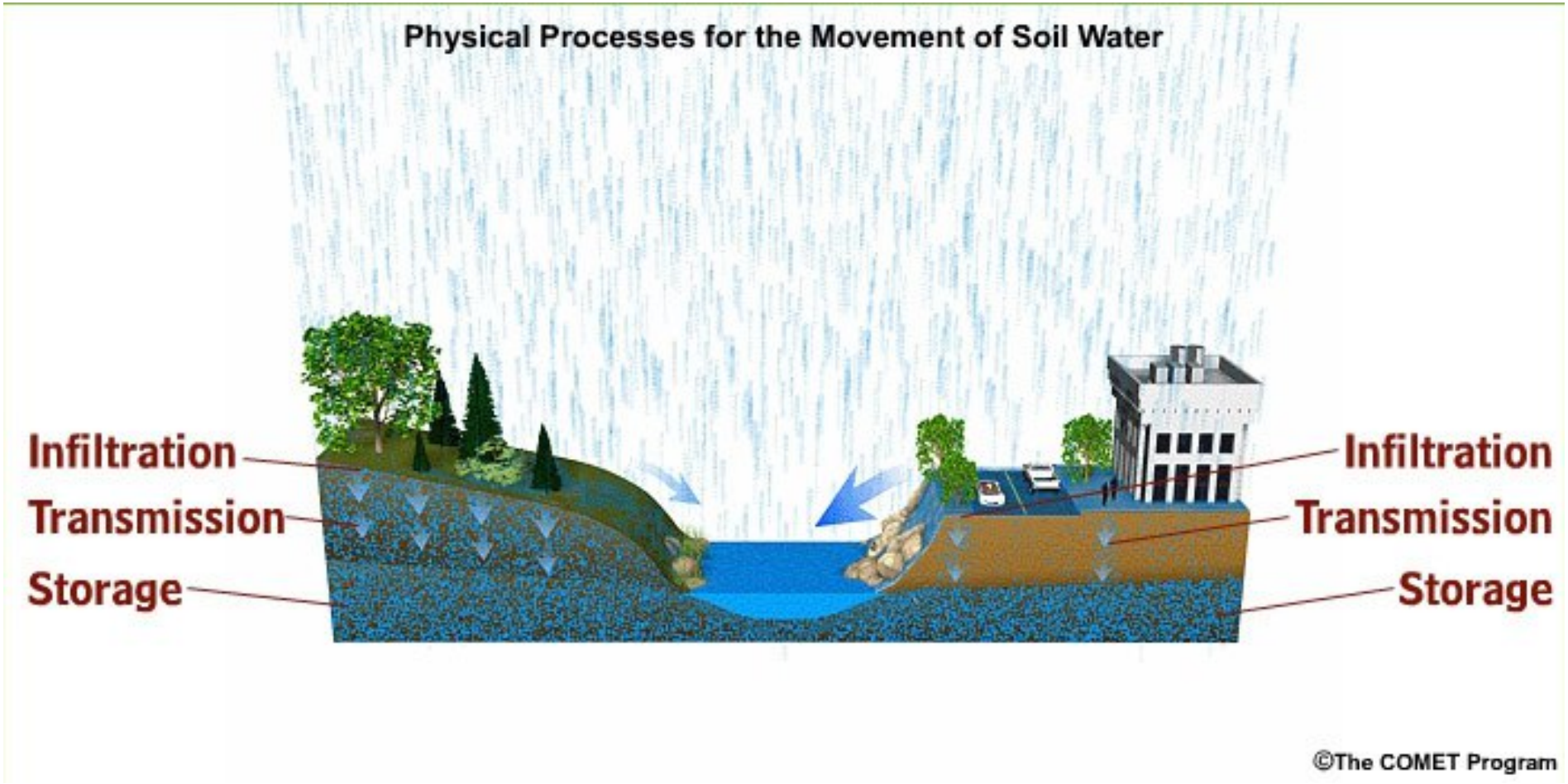
Difference between the two values will give Evapo-transpiration

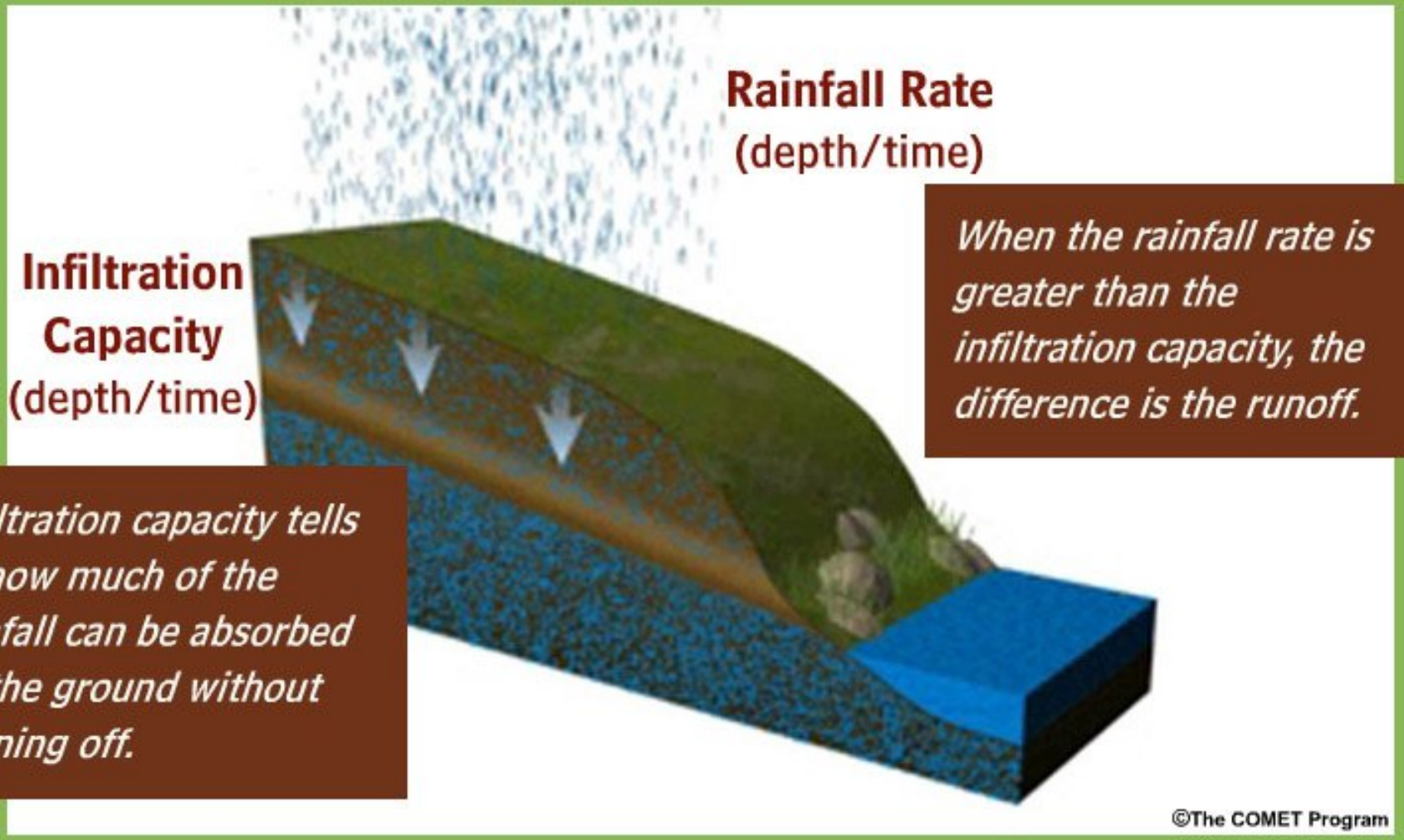
Measured
Water Added to
the field



Measured Infiltrated water
from the field

Infiltration




Rainfall Rate
(depth/time)
Infiltration Capacity
(depth/time)
When the rainfall rate is greater than the infiltration capacity, the difference is the runoff.
Infiltration capacity tells us how much of the rainfall can be absorbed by the ground without running off.
©The COMET Program

Infiltration

- **Infiltration** is the movement of water through the soil surface.
- **Infiltration Capacity** is the maximum rate at which a given soil at a given time can absorb water which is designated by f_c and expressed as cm/hr.

- The actual rate of infiltration f can be expressed as
- $f = f_c$ when $i \geq f_c$
- $f = i$ when $i < f_c$
- i = intensity of rainfall

Continued....

Factors affecting Infiltration

- Precipitation
- Soil type
- Soil water content
- Vegetation cover
- Ground slope.

Infiltration is measured by infiltrometer. There are two types

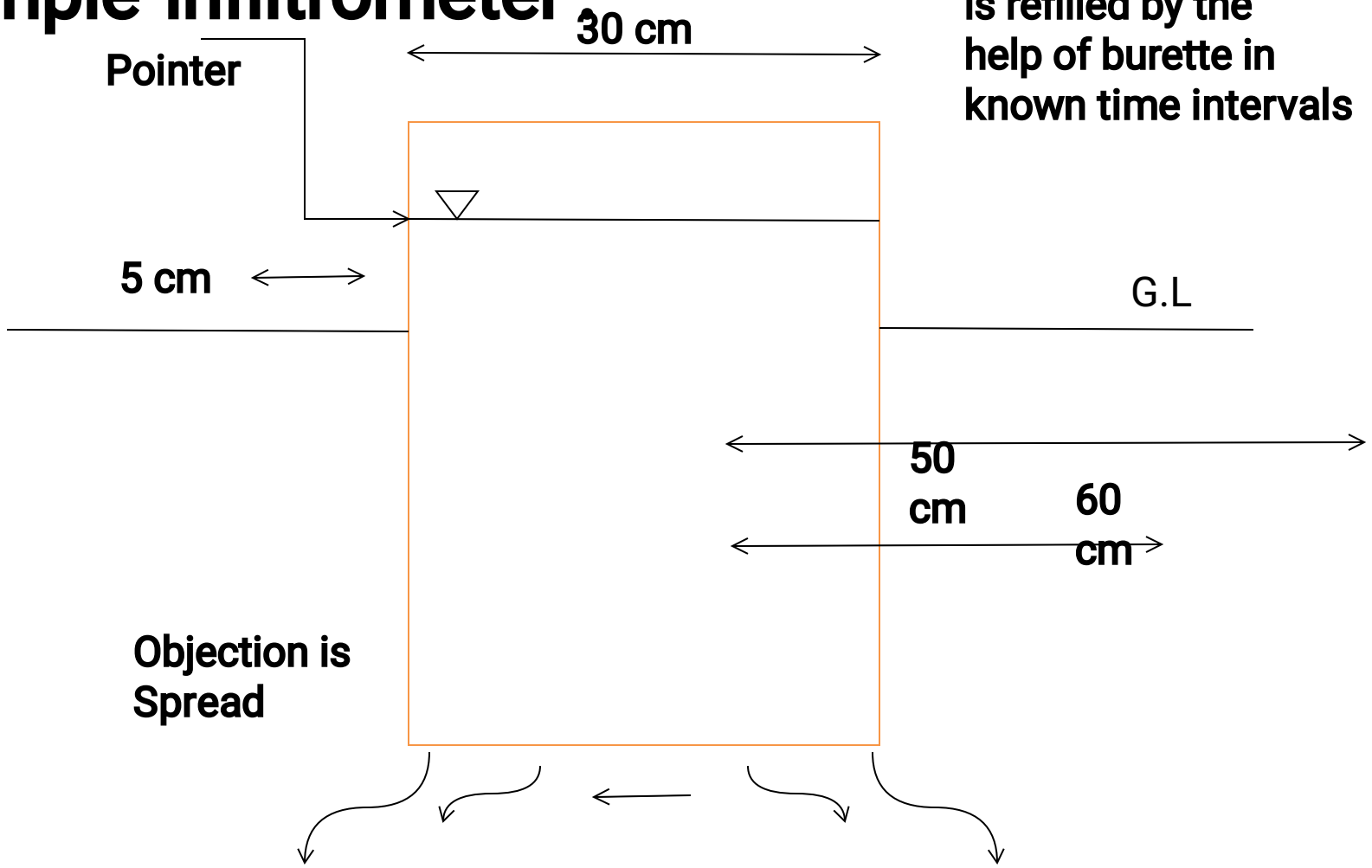
Flooding type

Simple infiltrometer

Ring infiltrometer

Measurement of Infiltration

Simple infiltrometer :



As the water level decreases, the ring is refilled by the help of burette in known time intervals

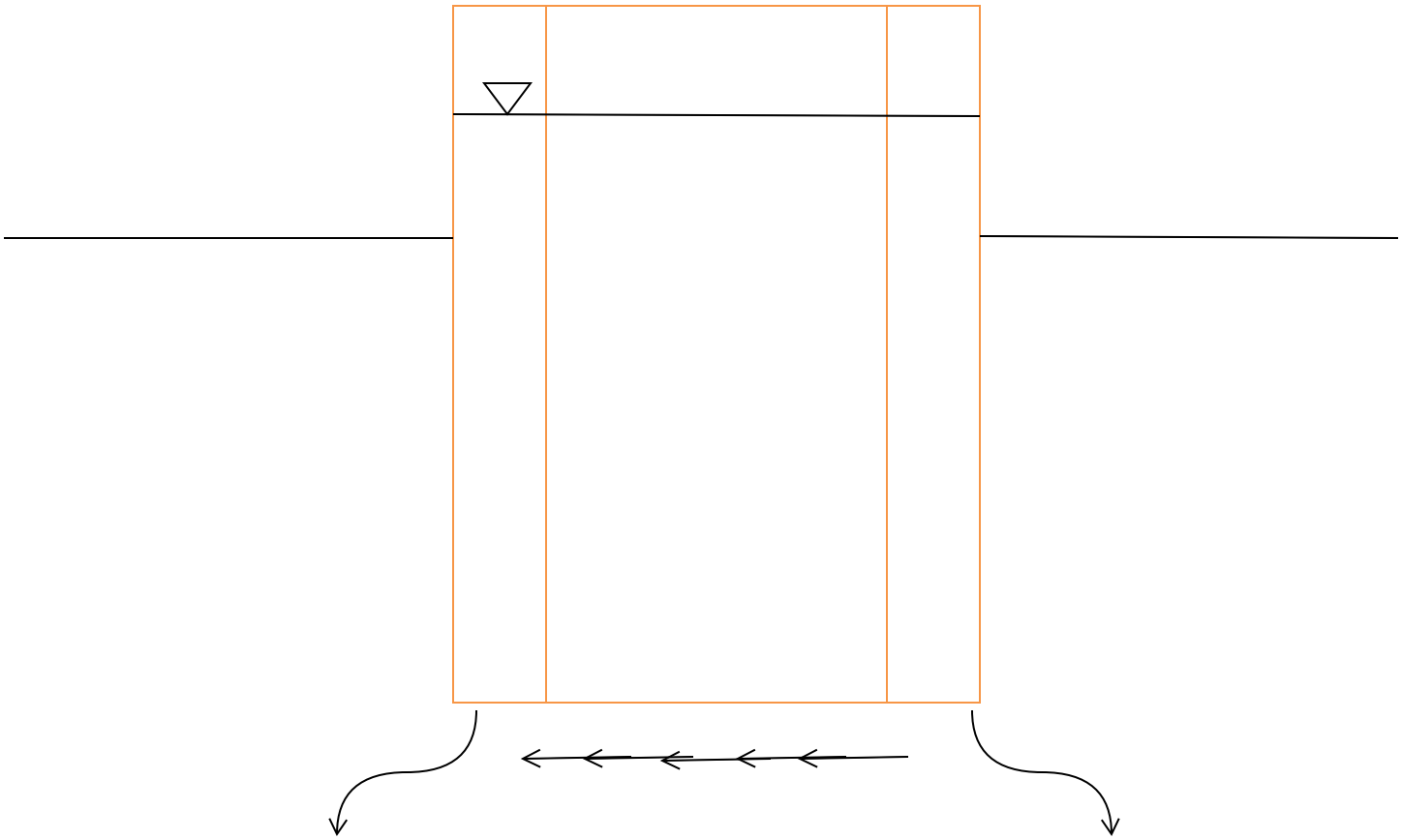
Double Ring Infiltrometer

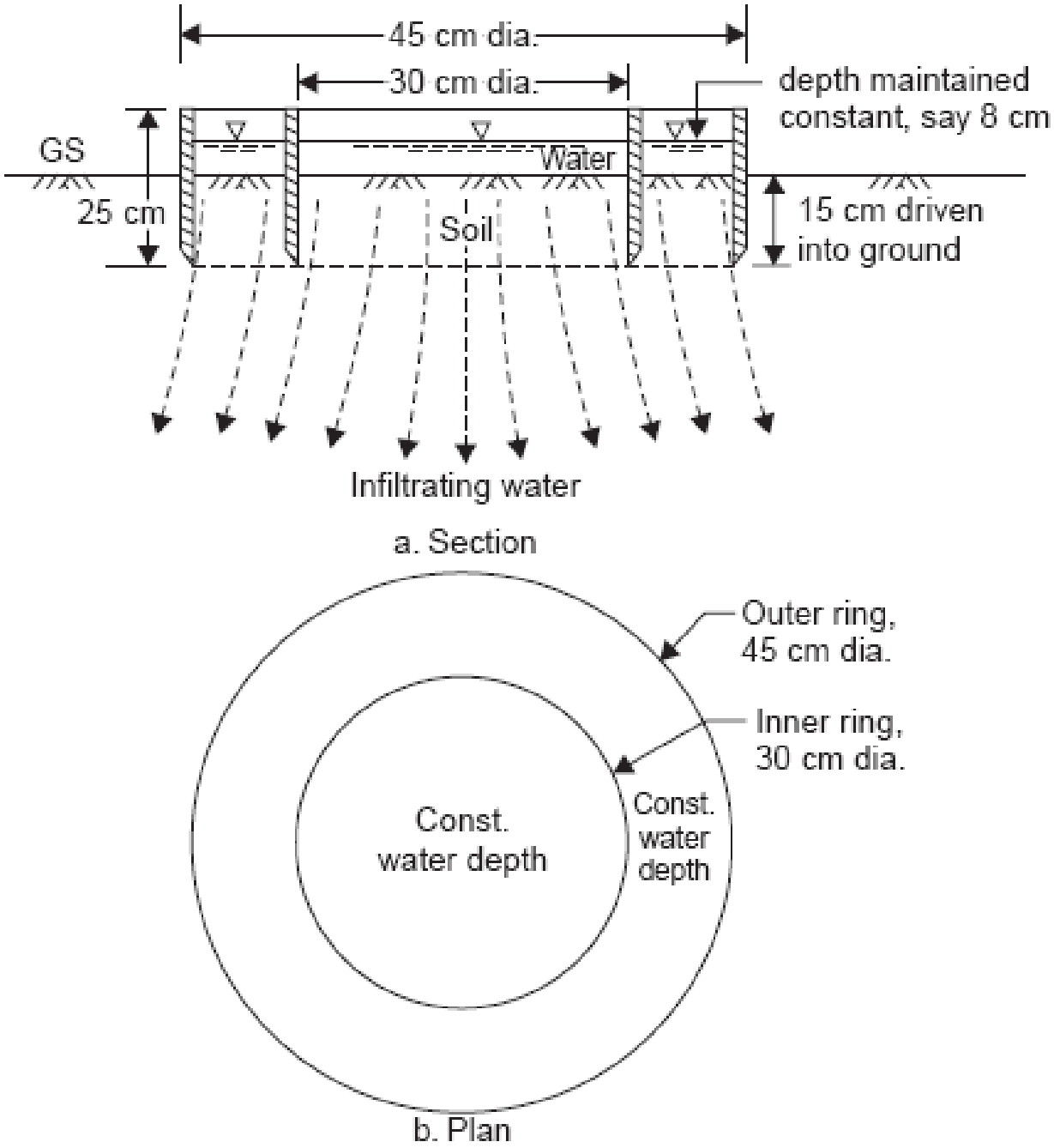
The two rings (22.5 to 90 cm diameter) are driven into the ground by a driving plate and hammer, to penetrate into the soil uniformly without tilt or undue disturbance of the soil surface to a depth of 15 cm.

- Point gauges are fixed in the centre of the rings and in the annular space between the two rings.

Measurement of Infiltration

Ring Infiltrometer :




45 cm dia.
30 cm dia.
depth maintained
constant, say 8 cm
GS
Water
25 cm
Soil
15 cm driven
into ground
Infiltrating water
a. Section
Outer ring,
45 cm dia.
Inner ring,
30 cm dia.
Const.
water depth
Const.
water
depth
b. Plan

- Water is poured into the rings to maintain the desired depth (2.5 to 15 cm with a minimum of 5 mm) and the water added to maintain the original constant depth at regular time intervals (after the commencement of the experiment) of 5, 10, 15, 20, 30, 40, 60 min, etc. up to a period of at least 6 hours is noted and the results are plotted as infiltration rate in cm/hr versus time in minutes.

- The purpose of the outer tube is to eliminate to some extent the edge effect of the surrounding drier soil.

Infiltration Capacity values

- Infiltration capacity for given soil decreases with time from the start of rainfall, i.e. it decreases with degree of saturation and depends on type of soil. Horton (1930) expressed the decay of infiltration capacity with time as
  - $f_{ct} = f_{cf} + (f_{co} - f_{cf})\, e^{-K_h t}$ for $0 \le t \le t_d$
- f_{ct} = *infiltration capacity at any time t from start of rainfall*
- f_{cf} = *final steady state value*
- f_{co} = *initial infiltration capacity at t = 0*
- *t = time from beginning of storm*
- t_d = *duration of rainfall*
- K_h = *constant depending on soil characteristics and vegetation*

Measurement of Infiltration

- Infiltration rate can be measured by simultaneous measurements of rainfall and runoff.
- Infiltration may be determined by using equations or infiltration indices.

Measurement of Infiltration Using Equations

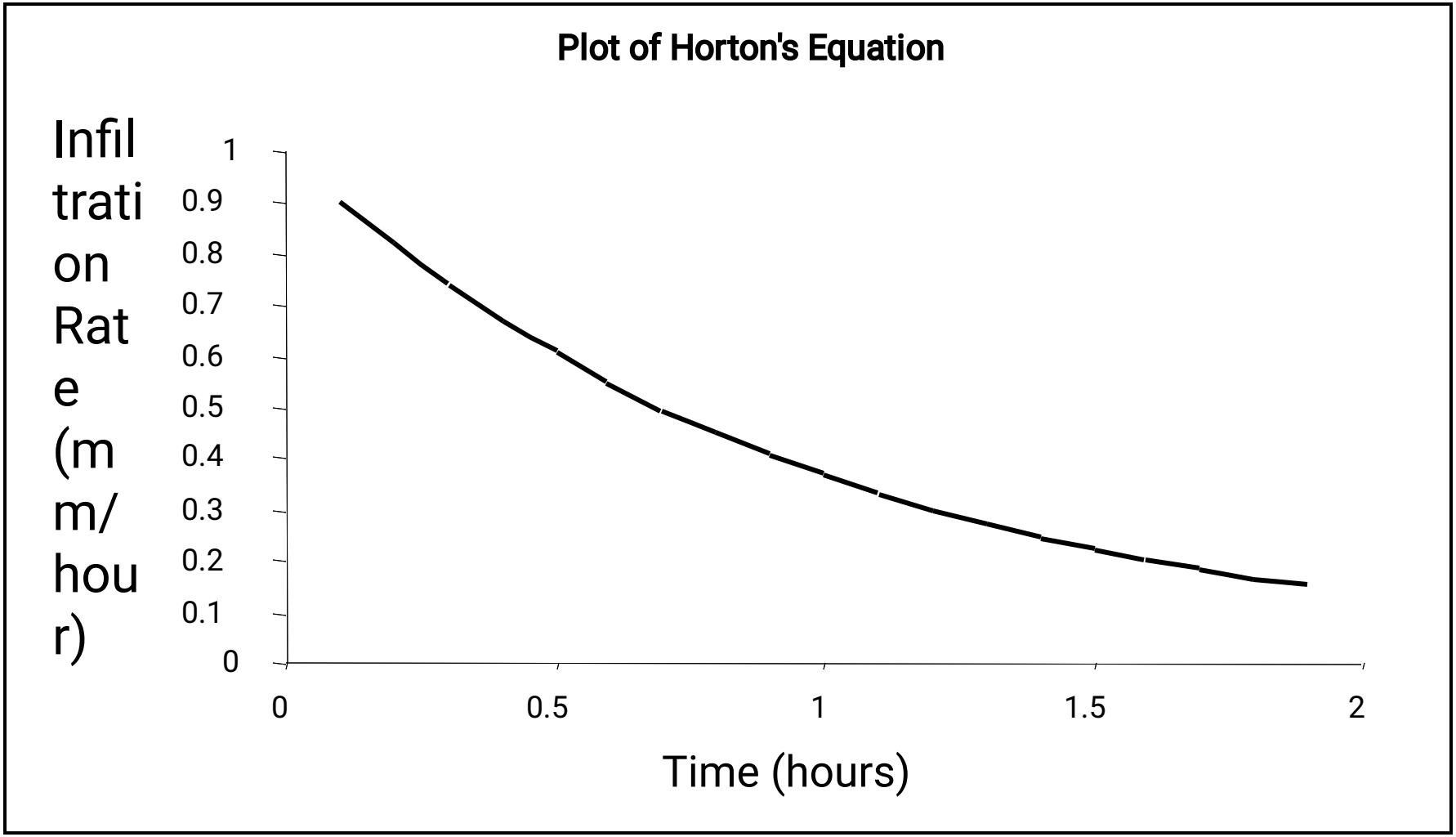


Fig. 4.4 Plot of Horton's Equation

Continued....

Infiltration Indices

- The average infiltration rate is called infiltration index.
- There are two types of indices which are common
  - Ø Index
  - W Index

Continued....

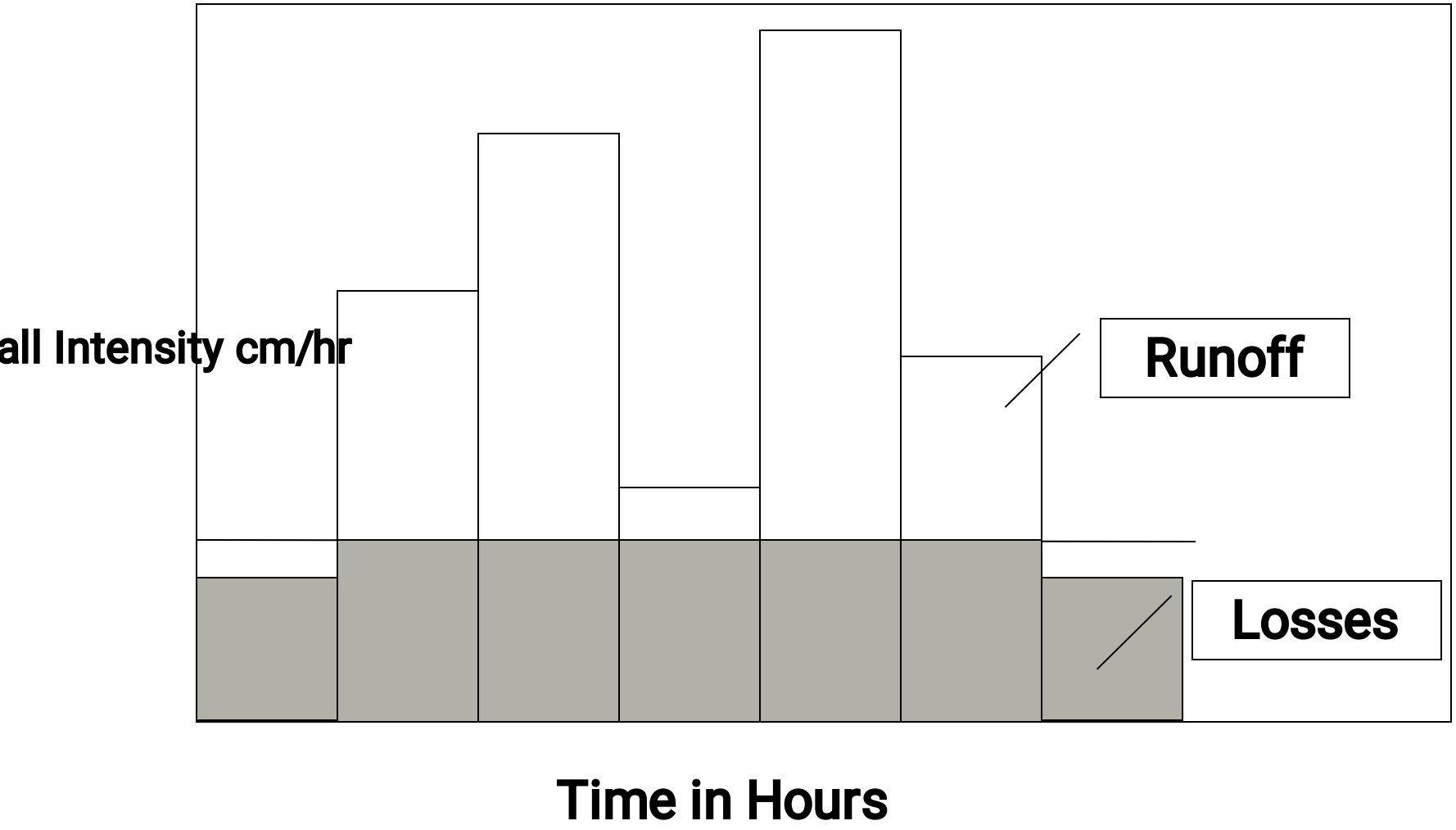
Infiltration Indices

- Index is average rainfall above which rainfall volume is equal to runoff volume. index is derived from rainfall hyetograph with edge of resulting runoff volume. value is found by treating it as a constant infiltration capacity.

- If rainfall intensity is less than then infiltration rate is equal to rainfall intensity. However, if rainfall intensity is larger than , the difference between rainfall and infiltration is an interval of time represents the runoff volume.

Continued....

Infiltration Indices



Continued....

W Index

- W index gives the average rate of infiltration. It can be found by subtracting the sum of total direct runoff, interception and depression storage from the total rainfall.

If, F = Total Infiltration

T = Duration of Storm

P = Depth of Precipitation

R = Runoff Depth

I = Interception and depression storage, then

$F = P - R - I$

And 'W' index = $F / T = (P - R - I) / T$

For I = 0, 'W' Index = ... Index

Thanks